Drill Machine

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 22 labs

Covers these lab items: Drill machine; Drill machine and torque wrench; Drill machine with drill bits of different sizes; Drill machines; Drilling Machine; Electric Drill Machine; Electrical hand drill machine; Electrical hand drill machine MII Class 1 Compliant; Power drills and drivers; Powered Hand drill

What it is used for

A hand-held electric drill machine makes clean round holes in wood, metal, plastic and (with a masonry bit and hammer mode) brick or concrete. It is used for fitting, fixing, assembly and repair work in the lab.

Main parts

- Body and handle with trigger switch
- Chuck and chuck key (holds the drill bit)
- Forward / reverse switch
- Speed control dial
- Hammer / drill mode selector (if available)
- Side handle for a firm two-hand grip
- Power cord and plug

Before you start

1. Check the cord, plug and body for cuts or cracks; never use a damaged machine
2. Select the correct bit: HSS for metal, wood bit for wood, masonry bit for brick/concrete
3. Fit the bit fully in the chuck and tighten with the chuck key; remove the key
4. Clamp the job firmly in a vice or with C-clamps
5. Mark the hole centre with a centre punch
6. Wear safety goggles; tie back long hair and remove loose clothing, gloves and jewellery

How to use

1. Plug in with the switch OFF
2. Hold the machine with both hands, at 90° to the surface
3. Place the bit tip on the punch mark
4. Start at low speed until the hole begins, then increase speed
5. Apply steady, light pressure; let the bit do the work
6. For deep holes, pull the bit out now and then to clear chips
7. Ease the pressure just before the bit breaks through
8. Release the trigger and let the bit stop fully before lifting the machine away
9. Switch off and unplug before changing the bit

⚠ Safety

- Never hold small jobs in your hand while drilling
- Keep fingers away from the rotating bit and chuck
- Do not touch the bit right after drilling; it is hot
- Use lubricant (cutting oil) when drilling metal
- Do not force the machine; if it jams, release the trigger at once
- Use only on a properly earthed socket; keep away from water

Care & storage

- Unplug and clean dust and chips with a brush after use
- Store bits in their case; replace blunt or bent bits
- Oil the chuck lightly now and then
- Store the machine in its box in a dry place

Common mistakes to avoid

- Leaving the chuck key in the chuck
- Drilling without clamping the job
- Using a wood bit on metal
- Pressing too hard, which breaks small bits